

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: REPRODUCTION IN PLANTS (SUB-STRAND 2.1)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit? Use your notebook, your labelled dissection diagrams, your fruit classification table and the class charts as evidence. Write your answers in full sentences in the space under each prompt. In Sections 2 to 4 use the Claim-Evidence-Reasoning (CER) structure. Use examples from our unit, such as cassava, sugarcane and sweet potato in Kisumu, Kisii and Siaya, and maize, beans, pawpaw, avocado and passion fruit. Draw neat, labelled diagrams where asked.

Part 1: Making Seedless New Plants

A farmer in Kisii plants cassava stem cuttings and a woman in Siaya plants sweet potato vines. Neither uses seeds. (a) Name three natural vegetative structures we examined in class (for example from sweet potato, onion, ginger, grass or bryophyllum) and say where on each a new plant can start. (b) Explain how a cassava cutting can become a whole plant without a flower, using the idea of cell division. (c) Describe how the new plants compare with the parent plant. (d) Give one advantage and one disadvantage of growing crops this way.

(a) Any three of these are correct. Sweet potato tuber: new shoots start from buds on the tuber, using the food stored in it. Onion bulb: new plants grow from buds inside the bulb, using the stored food in its fleshy leaves. Ginger rhizome: new shoots start from buds at the nodes. Grass runner: a new plant starts at the nodes along the runner. Bryophyllum: small plantlets start at the leaf margin. (b) The cutting is stem tissue from the parent plant. Its cells divide by mitosis and form new roots and shoots using food stored in the stem. No flower, seed or partner plant is needed because only one parent is involved. (c) The new plants are genetically identical to the parent. They are clones. (d) Advantage: growing is fast and the farmer keeps the good traits of the parent plant, with no pollination needed. Disadvantage: there is no variation, so one disease, such as banana wilt in Kisii and Meru, can affect all the plants in a field.

Part 2: Flowers, Pollination and the Lone Tree (CER)

A compound in Nairobi has a single pawpaw tree that is covered in flowers every year

Claim: The lone pawpaw tree does not fruit because it cannot receive compatible pollen from its own flowers, so it needs a partner tree to supply pollen. Evidence: In our station work with flower charts and wire/bead models, we saw that some trees have only male or only female flowers (dioecy), as in pawpaw. We also saw that pollen and stigma can be ready at different times (protandry),

but never sets fruit. Write a CER explanation. Claim: state why the tree does not fruit. Evidence: use at least two observations from our flower dissections, wire/bead model stations or animation work. Reasoning: explain how the features of flowers and trees that we studied at the model stations stop fruit from forming on a lone tree, and why a partner tree helps. Remember to separate pollination from fertilisation.	as in maize tassels and silks; protogyny), that stamens and styles can sit at different heights (heterostyly), and that some stigmas reject their own pollen (self-incompatibility, as in passion fruit). In the dissections we saw that fruit needs pollen from the stamens to reach the stigma of a carpel with an ovary. Reasoning: These features promote cross-pollination and give genetic variation. A lone male pawpaw tree has pollen but no ovary to form fruit, and a lone female tree has ovaries but no pollen of its own, so no compatible pollen reaches its stigmas. Where a tree does have both parts, timing, structure or chemical rejection can still prevent its own pollen from fertilising its ovules. Pollination is only the transfer of pollen. Without compatible pollen landing on a receptive stigma, no pollen tube grows and no fertilisation occurs, so the ovary does not develop into a fruit. A partner tree supplies compatible pollen carried by insects or wind, so the flowers can set fruit.
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Part 3: Double Fertilisation and Fruit Development

(a) Describe, in order, what happens from the time compatible pollen lands on a stigma until a seed begins to form. Name the structures the pollen tube passes through. (b) State what each of the two fusions produces and what each product does for the seed. (c) Say what the ovules and the ovary become after fertilisation. (d) Using a fruit you sectioned in class (tomato, orange, bean pod, passion fruit or cucumber), name its placentation type and explain how its seeds show the link to fertilised ovules.	(a) The pollen grain lands on a receptive stigma and germinates. A pollen tube grows down the style to the ovule and enters through the micropyle. It releases two sperm nuclei into the embryo sac. (b) One sperm nucleus fuses with the egg cell to form the zygote, which becomes the embryo. The other sperm nucleus fuses with the two polar nuclei to form the triploid endosperm, the food store. This is double fertilisation. The endosperm is what we eat in a maize grain used for ugali. (c) The fertilised ovules become seeds and the ovary becomes the fruit. (d) Any one sectioned fruit is enough. The tomato and the orange have axile placentation, with seeds attached to a central axis in separate chambers. The bean pod has marginal placentation, with seeds attached along one edge. Passion fruit and cucumber have parietal placentation, with seeds attached to the wall. The seeds we found in the fruit are the fertilised ovules, attached in the pattern set by the ovary. (Free-central placentation, with seeds on a central column, was not sectioned in class; we studied it from a cached image or example.)
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Part 4: Comparing Sexual and Asexual Reproduction in Kenyan Farming (CER)

A farmer asks you whether it is better to plant cassava from cuttings or maize from seed, and what would happen if a new disease arrived in each field.	Claim: Asexual reproduction gives fast, uniform plants, while sexual reproduction gives variation that helps a population survive change. Neither is better in all conditions. Evidence: Cassava, sugarcane in Kisumu and Nyando, sweet potato and banana grown from cuttings, setts or suckers are identical to the parent and spread quickly, but one disease such as banana wilt or cassava mosaic disease can affect all of them. Maize, beans and passion fruit grown from seed vary from plant to plant. A lone pawpaw or avocado needs a partner tree for pollination before it can fruit. Reasoning: Asexual reproduction needs only one parent and uses
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Write a CER response. Claim: state the trade-off between the two strategies. Evidence: use at least two case examples from our case cards or class work. Reasoning: link the number of parents, the type of cell division or fertilisation, and the amount of variation to survival in changing conditions. Finish by explaining why neither strategy is always best.	mitosis, so offspring are clones with the same strengths and weaknesses. This is useful for keeping good traits and covering a field quickly, but a disease that attacks one plant can attack all of them. Sexual reproduction needs pollination and double fertilisation, which mix the genes of two parents. The variation means some offspring may resist a disease or cope with changing conditions. The costs are that it needs flowers, a pollen source and more time. The best choice depends on the environment and the farmer's goal.
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Part 5: Revised Model and Annotated Diagram

Look back at the first model you drew in Lesson 1, comparing a sweet potato making a new vine with a bean seed making a plant. (a) Draw a revised annotated diagram that explains how a cassava cutting, a maize or bean plant and a lone pawpaw tree each end up with, or without, new plants or fruit. Label the main structures and processes. (b) Write three to four sentences describing how your thinking changed and one question you still have.	(a) A good diagram has two pathways. Asexual pathway: a cassava or sweet potato cutting or tuber with buds or nodes, mitosis, new roots and shoots, and a clone of the parent, with no flower or seed. Sexual pathway: a flower showing anther, filament, stigma, style, ovary and ovule. Pollen is transferred, with an arrow for cross-pollination from a partner pawpaw tree. A pollen tube passes through the style to the micropyle. Sperm nucleus 1 + egg gives the zygote. Sperm nucleus 2 + two polar nuclei gives the triploid endosperm. Ovule becomes seed and ovary becomes fruit, with placentation labelled. A note beside the lone pawpaw shows that barriers such as dioecy or self-incompatibility prevent pollination, so no fertilisation and no fruit. (b) Sample answer: At first I thought all plants made new plants from seeds, and that a flowering tree must give fruit. Now I know that cuttings use mitosis to make clones without flowers. I also know that fruit needs pollination and double fertilisation, and that barriers to self-pollination mean a lone tree may need a partner. I still wonder how farmers can choose between cuttings and seeds when both disease risk and yield matter.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of the phenomenon	Gives a clear, accurate claim that answers the driving question for both cassava-type and pawpaw/avocado-type plants, using Kenyan examples from the unit.	Gives a mostly accurate claim that covers both cases, with limited use of examples.	Claim is unclear, covers only one case, or contains misconceptions, for example that flowers alone guarantee fruit.
Evidence from investigations	Uses at least two specific observations from dissections, stations, animations, fruit sections or case cards, accurately	Uses one or two observations, with some detail missing.	Gives little or no evidence, or evidence that is vague or not from our lessons.

	and with detail.		
Scientific reasoning and terminology	Correctly links mitosis and cloning, barriers to self-pollination, pollination versus fertilisation, double fertilisation (zygote and triploid endosperm), and ovule-to-seed and ovary-to-fruit development.	Links most ideas correctly, with minor errors or missing links.	Reasoning is incomplete, or confuses pollination with fertilisation or the products of the two fusions.
Comparison of sexual and asexual reproduction	Explains advantages and disadvantages of both strategies, including variation and disease risk, and why the best choice depends on the environment.	Names advantages and disadvantages of both strategies with limited explanation.	Describes only one strategy or lists points without explanation.
Annotated diagram and presentation	The diagram is neat and accurate, with all key structures and processes labelled for both pathways, and the written reflection is clear.	The diagram has most labels correct, with a few omissions, and the reflection is brief.	The diagram is missing labels or has errors, or the reflection is missing.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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